SAINT MICHAEL COLLEGE OF CARAGA

THE IMPACT OF ADOPTING A POINT OF SALE SYSTEM ON OPERATIONAL EFFECTIVENESS AND EFFICIENCY.

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The increasing shift toward digital financial technologies has transformed the way businesses manage transactions, with Point-of-Sale (POS) systems and electronic payment platforms becoming essential tools for operational efficiency worldwide. This global trend has also influenced the Philippines, where businesses are progressively adopting digital solutions to enhance service delivery, financial transparency, and operational management. In the municipalities of Buenavista, Nasipit, and Carmen, this movement is evident through the growing use of the Alliance OS POS system, which enables local establishments, particularly in the food and beverage sector, to provide faster customer service, access real-time business information, and achieve greater accuracy in financial operations. The adoption of such technologies illustrates how international digital trends are reshaping business practices at the local level.

Globally, recent studies highlight the rapid evolution of e-wallets and digital payments, emphasizing their role in reshaping financial behavior and improving transactional experiences. In Indonesia, internet growth and financial technology have spurred widespread adoption of e-wallets, although research largely focuses on their benefits rather than potential risks (Underdown & Tamara, 2025). Internationally, POS systems have advanced through big data integration, enabling businesses to monitor inventory and restocking more efficiently (Shaba, 2024). In Nigeria, POS service providers have positively influenced agricultural markets by

increasing access to financial services for farmers (Dewi et al., 2021; Kholod et al., 2024; Lesiak et al., 2024; SHARKAWI, 2024). Studies also indicate that paperless billing improves payment speed, security, and overall efficiency (Faruki et al., 2023; Swathi et al., 2020), while electronic financial systems supported by internet and mobile technologies enhance affordability and accessibility (Mahathir et al., 2025; Twaliwi et al., 2023).

In the Philippines, digital payment adoption is increasing but remains partial, influenced by policy improvements, digital infrastructure, and user trust (Calderon, 2025; Glindro et al., 2024; Mae Jacob, 2024; Sentral Ng Pilipinas, 2024). Mobile wallets such as GCash remain the most widely used cashless payment method due to convenience and security, followed by debit/credit cards and online banking, all of which contribute to the growing trust in digital financial systems (Borja et al., 2024; Hernandez et al., 2022; Mathew Jun P. Mariani, 2022; Ramos de Luna et al., 2024). Institutions such as Mariano Marcos State University have implemented self-service kiosks to reduce queues and enhance customer experience, reflecting local adaptation to digital technologies (Calderon, 2025; Marie Hilay et al., 2021; Timoteo, 2022). Despite this progress, several studies highlight recurring issues such as inventory mismatches and inaccurate sales reporting, reinforcing the need for stronger internal controls and dependable POS systems in businesses. These insights bridge the global technological context with the Philippine setting, emphasizing the importance of improving POS efficiency and accuracy in the country's food and beverage sector.

With the increasing adoption of digital payment systems in the Philippines, local food and beverage establishments continue to face challenges in accuracy, inventory management, and

service quality. Research indicates that point-of-sale (POS) systems can enhance operational efficiency by reducing order errors, improving inventory tracking, and speeding up service, all of which contribute to higher customer satisfaction (Kumar & Saini, 2021; Chawla & Joshi, 2019; Lin & Chen, 2020). Studies also show that POS systems facilitate real-time monitoring of sales and stock, supporting more informed managerial decisions and minimizing operational losses (Ahmed et al., 2022; Tan et al., 2018). Despite this evidence, the effectiveness of POS systems in small to medium-sized Philippine hospitality businesses remains underexplored. This study addresses this gap by examining the use of POS systems in establishments in Buenavista, Nasipit, and Carmen, aiming to identify the demographic profile of respondents (age, sex, educational attainment, work experience), assess the impact of POS systems on order accuracy, service speed, and customer satisfaction, and determine the prevalence of POS usage among local businesses. By integrating empirical findings with local operational data, the study seeks to evaluate how the Alliance OS POS system can enhance efficiency, reduce errors, and improve decision-making in Philippine F&B establishments.

The findings of this study will be valuable to multiple stakeholders within the food and beverage industry. Restaurant owners and managers can use the results to make informed decisions about acquiring or upgrading POS systems to reduce human error, streamline operations, and increase profitability. Employees will benefit from more efficient workflows and reduced workload related to manual tracking and order processing, potentially improving service quality and job satisfaction. Customers will experience faster, more accurate service, contributing to higher satisfaction and loyalty. For the Bachelor of Science in Hospitality and Tourism

Management program at Saint Michael College of Caraga, this study can support curriculum enhancement, research, and practical training by providing relevant insights into the use of POS systems in hospitality operations. The findings may be used in classroom discussions, case studies, and hands-on activities to strengthen students' understanding of restaurant management, digital transactions, and operational efficiency. Additionally, the study's proposed recommendations or models for effective POS system usage can help prepare students for data-driven and technology-oriented practices in the hospitality and tourism industry.

1.2 Theoretical Framework

This study is anchored in the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2), which explains how individuals and organizations accept and use new technologies. UTAUT2 posits that technology adoption is influenced by key determinants: performance expectancy, effort expectancy, social influence, facilitating conditions, price value, and habit. In the context of food and beverage (F&B) establishments adopting point-of-sale (POS) systems, performance expectancy reflects the extent to which managers and staff perceive the system as improving operational efficiency and customer service. Effort expectancy relates to the perceived ease of learning and using the POS system. Social influence captures the impact of peers, industry norms, or managerial encouragement on adoption decisions. Facilitating conditions involve the availability of supporting infrastructure, such as internet access, system security, and reliable technical support. Price value assesses whether the perceived benefits of the POS system outweigh its costs. Habit considers the extent to which prior experiences with technology shape the likelihood of adopting and consistently using the system.

Once the POS system is adopted, it generates several operational outcomes that align with the dependent variables of the study, including improved awareness, convenience, security and safety, trust, reduced reliance on cash, social pressure to modernize and increased customer spending. These outcomes directly contribute to enhanced operational effectiveness and efficiency by streamlining transaction processes, reducing errors, improving data accuracy, strengthening financial control and enabling faster, more reliable customer service. Through this theoretical lens, POS adoption serves as the mediating mechanism that connects establishment characteristics and system conditions to measurable operational gains.

This theoretical framework is significant because it provides a structured explanation of how and why POS systems influence organizational performance. It not only identifies the factors that drive adoption but also clarifies the pathways through which the technology generates operational benefits. By grounding the study in UTAUT2, the research ensures that the relationship between the independent variables, POS adoption and operational outcomes is logically connected, theoretically supported and aligned with existing technology adoption literature. This enables a deeper and more systematic examination of the impact of POS systems on operational effectiveness and efficiency.

1.3 Review of Related Literature

Point-of-Sale (POS) systems are widely recognized for enhancing operational effectiveness across retail, hospitality, and service sectors. Research consistently shows that POS systems streamline transaction processes, reduce reliance on manual labor, and minimize human error, collectively leading to faster service delivery and improved customer satisfaction (Alt, 2021;

Azhari & Sutarman, 2024; Widjaja et al., 2021). Beyond these immediate operational benefits, studies reveal a broader pattern: the automation of sales recording and real-time inventory tracking not only strengthens internal controls and prevents stockouts but also enables more accurate forecasting and strategic decision-making. This integration of efficiency, error reduction, and data-driven management suggests that POS systems contribute not merely to isolated improvements but to a systemic enhancement of productivity and resource management within organizations.

Research further emphasizes that POS analytics strengthen managerial functions such as supplier coordination, demand planning, promotional strategies, and loss prevention, ultimately improving profitability. Modern POS features—such as digital payments, e-receipts, and loyalty programs—also enhance customer experience and retention (Hassan et al., 2021; Jayson Diaz1 et al., 2022; Naik et al., 2024).

However, several studies note challenges in adoption, especially among SMEs, including high upfront costs, maintenance expenses, financial risks, and concerns regarding data security, privacy, and fraud (Akinsuru Adedoyin & Oreoluwa Ojo, 2024; Mamudu Zebedee Udo, 2021; Rubyet Hossain; et al., 2020). Limited technological infrastructure, lack of digital literacy, and insufficient organizational readiness also hinder effective use of POS data for operational improvement. Despite these barriers, the overall evidence demonstrates that adopting POS systems leads to more streamlined operations, better resource management, and enhanced service efficiency, making it a vital tool for businesses aiming to remain competitive and operationally effective.

Establishments' profile

Demographic factors such as age, gender, employment status, and educational attainment play a significant role in shaping individual behaviors and adoption patterns in various contexts, including technology use and financial practices. Age often influences familiarity and comfort with new technologies, with younger individuals typically demonstrating higher adaptability, while gender can affect preferences and engagement levels in adopting innovations. Employment status determines access to financial resources and the perceived necessity of adopting new systems, whereas educational attainment contributes to awareness, understanding, and confidence in using technological solutions. These variables collectively provide a comprehensive framework for understanding behavioral intentions and adoption patterns, as highlighted in studies examining technology adoption, including e-wallet usage (Rahi et al., 2019).

Identified perceived benefits, effort expectancy, social influence, trust, awareness, and demographic variables—including age, gender, and educational attainment—as significant predictors of technology adoption. These findings highlight the critical role of both psychological and demographic factors in shaping organizational decisions to implement technological innovations. In the context of point of sales (POS) systems, these insights suggest that understanding these determinants is vital for evaluating how the adoption of such systems can enhance operational processes, optimize transaction management, and improve overall organizational efficiency. By considering both individual and contextual factors, organizations can better facilitate the successful integration of POS technologies, thereby maximizing their impact on operational effectiveness Gholami et al. (2010).

The store crew, acting as cashier-level staff, are the primary front-line users of the POS system, operating it during daily transactions and sales activities. In addition, managerial personnel use the POS system for supervisory functions, such as monitoring operations, reviewing inventory data, and making management decisions. This indicates that the POS system is role-based, with its use concentrated specifically on cashiers and managers within the establishment (Bienvenido, 2023)

Customers' perception

Research on customers' perception of Point-of-Sale (POS) systems in Nigeria highlights the importance of various technological, operational, and service-related variables that shape user experience within cashless and e-payment environments. Key factors frequently examined in the literature include Internet quality, which influences connectivity and transaction speed; Completeness and Consistency, which relate to the reliability and uniformity of POS functions; Observability, referring to the visibility and clarity of transaction processes; and Accuracy, which ensures error-free operations during payments. Additional variables such as Location, Pricing, and Security are also recognized as important contextual and structural elements that affect customer interaction with POS machines. Together, these variables provide a comprehensive framework for understanding how customers form perceptions of POS technology and how these perceptions can guide improvements in Nigeria's cashless and electronic payment systems (Jimoh et al., 2023).

This chapter reviews literature on the adoption and impact of POS systems on operational effectiveness and efficiency, focusing on key factors influencing usage.

Previous studies and survey results indicate a positive and statistically significant relationship between the variables of interest, confirming their relevance in the context of the study. The observed moderate correlations suggest that each variable maintains a degree of independence, allowing for meaningful inferences in predictive modeling. Specifically, the determinants of electronic payment systems, namely awareness, convenience, safety and security, trust, and social influence, demonstrate a positive association with consumers' spending behavior in Nigeria. This underscores the critical role these factors play in shaping consumer adoption and usage of electronic payment systems within the Nigerian context (Oyelami et al. 2020).

Age significantly affects the adoption of digital payment systems. Al-Qudah, Al-Okaily, Shiyyab, et al. (2024) found that younger users are more likely to adopt POS systems because of their higher digital literacy and perception of convenience and efficiency. In contrast, older adults tend to rely on traditional payment methods due to limited familiarity with digital technologies.

Similarly, Franco (2023) reported that younger consumers show greater openness to using digital payment platforms, while older users often experience difficulties related to usability and confidence. Jiang and Lau (2023) further explained that older adults' adoption is hindered by security concerns and fear of fraud, whereas younger users are more trusting of digital systems.

Martinez and McAndrews (2022) emphasized that user-friendly system designs and targeted educational efforts can help increase digital payment adoption among older adults. Overall, the literature confirms that age plays a significant role in digital payment adoption, with younger users showing higher acceptance than older users.

Gender has been found to influence the perception and use of POS technologies. Gutierrez-Leefmans and Olaleye (2021) reported that male users tend to prioritize efficiency, speed, and functional performance when using digital payment systems. Their study suggests that men are more likely to adopt POS technologies when these systems enhance transaction convenience.

In contrast, Malarvizhi et al. (2022) found that female users place greater importance on trust, security, and reliability in digital payment platforms. Concerns related to data privacy and transaction safety strongly influence their willingness to use POS systems. Reynolds et al. (2023) emphasized that social influence and perceived approval from peers play a significant role in shaping women's acceptance of POS technologies.

Educational attainment also plays a crucial role in the adoption of digital payment systems. Al-Qudah, Al-Okaily, Alqudah, et al. (2024) found that individuals with higher educational attainment demonstrate greater confidence and competence in using POS technologies, which leads to smoother transaction processes and higher adoption rates.

Franco (2023) emphasized that educated users are more capable of understanding the features and benefits of digital payment systems, allowing them to adapt more easily to technological changes. Hsiao and Tang (2025) further noted that higher levels of education are associated with stronger problem-solving skills and technological awareness, which positively influence users' willingness to adopt and continuously use digital payment platforms.

Employment status significantly affects POS efficiency, particularly in terms of employee training and adaptability. Kholod et al. (2024) found that well-trained employees are better able to

operate POS systems efficiently, resulting in faster transactions, improved accuracy, and reduced operational errors. Their study also emphasized that integrating POS systems with inventory and workflow management enhances overall operational performance and supports better demand forecasting. Yoon (2023) highlighted that organizations with adaptable and technologically competent staff experience smoother POS operations. However, the study noted that challenges persist when temporary or less tech-savvy employees are involved, as insufficient training can lead to slower processing times and system misuse.

Awareness User education and training are critical for successful POS adoption, particularly in underbanked regions. Goel et al. (2020) emphasized that providing clear guidance and demonstrations helps users understand system operations, increasing their willingness to adopt digital payment technologies. Isharyani et al. (2024) found that step-by-step onboarding programs reduce user errors and improve confidence when using POS systems. Ly and Ly (2024) also highlighted that structured training and practical demonstrations encourage consistent usage and facilitate smoother transactions, especially among first-time users. technology acceptance underscores that awareness of the functions and benefits of a system is a key driver of its adoption. Awareness here comprises not only recognizing a technology's existence, but also understanding its use cases and perceived usefulness, constructs central to foundational acceptance models like the Technology Acceptance Model (TAM) (Boonyong et.al, 2025).

Convenience System performance, including speed, ease of use, and perceived utility, directly affects user satisfaction and continued POS usage. Eren (2024) emphasized that fast and reliable transactions increase customer satisfaction and encourage repeated use of POS systems.

Mamudu Zebedee Udo (2021) found that users are more likely to adopt and continue using POS technologies when the system is easy to navigate and performs consistently without errors. Moncada et al. (2025) highlighted that operational efficiency—achieved through quick and accurate transactions, enhances overall user experience and supports long-term adoption of POS systems.

Security and safety Security concerns, such as fraud and data privacy, remain major barriers to POS adoption. Cinar and Kara (2023) emphasized that fear of unauthorized transactions and cyber threats can discourage users from adopting digital payment systems. Rawlings Fayeofori Fiberesima (2025) found that implementing technical safeguards, such as encryption and secure authentication, significantly improves user confidence in POS technologies. Santos and Bacalhau (2023) highlighted that educating users about security measures and safe usage practices enhances trust, reduces perceived risks, and encourages wider adoption of POS systems. Users assess the security of an information system by evaluating its ability to protect sensitive data and prevent unauthorized access; when users perceive adequate security, their trust in the system increases and encourages continued use (Abedisi et.al, 2025).

Trust in technology and institutions encourages adoption and long-term use of POS systems. Alrawad et al. (2023) found that users are more likely to adopt POS technologies when platforms are reliable and backed by reputable providers. Talwar et al. (2020) emphasized that strong brand reputation and consistent service quality enhance user trust, which in turn increases the frequency of POS usage. Zhang and Kim (2025) highlighted that trust-building measures, such as transparent policies and dependable support, positively influence user loyalty and continued

engagement with digital payment systems. studies on electronic payment technologies have established that security and trust are central determinants of user acceptance. Research on electronic payment and e-commerce systems shows that perceptions of transaction security strongly influence users' trust in a system, and this trust, in turn, affects their decision to adopt and use the technology (Adebisi et.al, 2025)

Social Influence plays a significant role in the adoption of POS systems, particularly in collectivist cultures. Asrol (2024) noted that users are more likely to adopt digital payment technologies when peers or family members recommend or regularly use them. Cao et al. (2024) emphasized that societal norms and expectations strongly shape individuals' willingness to engage with POS systems, as acceptance within a social group reinforces usage. Shah and Asghar (2023) highlighted that social pressure and peer approval can increase both initial adoption and continued use of POS technologies. Research on technology adoption consistently identifies social influence as a critical determinant of user intention and behaviour. Social influence refers to the extent to which individuals perceive that important others believe they should use a technology (Okoli et.al, 2025).

Limit of cash movements is accelerated by POS adoption, facilitating faster and cashless transactions. Glowka et al. (2023) emphasized that POS systems reduce reliance on physical cash, improving banking visibility and enabling more efficient financial monitoring. Islam et al. (2024) highlighted that POS adoption enhances operational efficiency by streamlining transactions, minimizing errors, and supporting smoother financial management for businesses and consumers.

POS adoption influences consumer spending behaviors, often resulting in higher transaction values and increased business revenue. Broekhoff and van der Cruijsen (2024) noted that the convenience and speed of digital payments encourage customers to spend more frequently and in larger amounts. Hidayati et al. (2023) emphasized that while POS systems boost sales, careful inventory and operational management are essential to handle increased transaction volumes efficiently. Ma et al. (2024) highlighted that integrating POS systems with inventory and workflow processes helps businesses capitalize on higher consumer spending while maintaining operational control.

Location

Location significantly influences the operational effectiveness and efficiency of businesses adopting a Point-of-Sale (POS) system. Kholod et al. (2024) emphasized that integrating POS data with location-tracking technologies, such as RFID, enables businesses to monitor customer movement patterns and transaction points within specific areas, supporting optimized store layouts, strategic product placement, and efficient staff allocation.

Similarly, Ighomereho & Omoyele (2022) highlighted that the reliability, ease of use, and fulfillment of POS systems across different service points are critical for ensuring smooth, secure, and efficient transactions, directly affecting operational performance. These findings indicate that strategic placement of POS terminals within specific locations enhances transaction efficiency, optimizes resource allocation, and improves overall customer satisfaction, demonstrating the importance of location in operational outcomes.

Internet

Internet connectivity is a vital factor influencing how effectively and efficiently a Point-of-Sale (POS) system operates. Walter (2024) argues that reliable multi-network cellular connectivity (e.g., switching between 4G/5G and Wi-Fi) ensures POS devices remain online even when one network fails, reducing transaction downtime and improving resiliency. Meanwhile, Kayla (2025) explains that slow or unstable broadband can severely delay payment processing, disrupt real-time inventory updates, and cause synchronization errors in cloud-based POS platforms, ultimately degrading both customer experience and operational throughput.

Consistency

Consistency in service delivery is a key factor in determining how well a POS system enhances operational effectiveness and efficiency. Gao & Fan (2021) found that when customers perceive their online and offline experiences to be congruent (i.e., consistent), satisfaction, repurchase intention, and word-of-mouth improve, whereas any discrepancy between channels undermines these outcomes.

In banking contexts, Olukemi Oluyemisi & Abba (2024) noted that the reliability dimension of POS terminal service, closely linked to consistent availability across POS channels, strongly influences customer satisfaction. Thus, ensuring consistent POS service quality across all touchpoints supports smoother operations, reduces customer frustration, and promotes greater operational efficiency.

Accuracy

Accuracy is a critical dimension in how adopting a POS system affects operational effectiveness and efficiencyOkofu (2025) developed an enhanced POS system for retail businesses in developing countries that ensures transactions are “reliable, accurate, faster, and secure” by using barcode scanning and a recommender-system architecture, minimizing human error and improving the precision of sales records.

Moreover, Zídková et al. (2025) discusses the digital registration of cash sales via POS systems in public finance, showing that accurate real-time reporting of cash transactions can strengthen fiscal resilience and operational transparency. Together, these studies highlight that accuracy in POS transactions not only reduces errors and discrepancies but also enhances data reliability for decision-making and operational control.

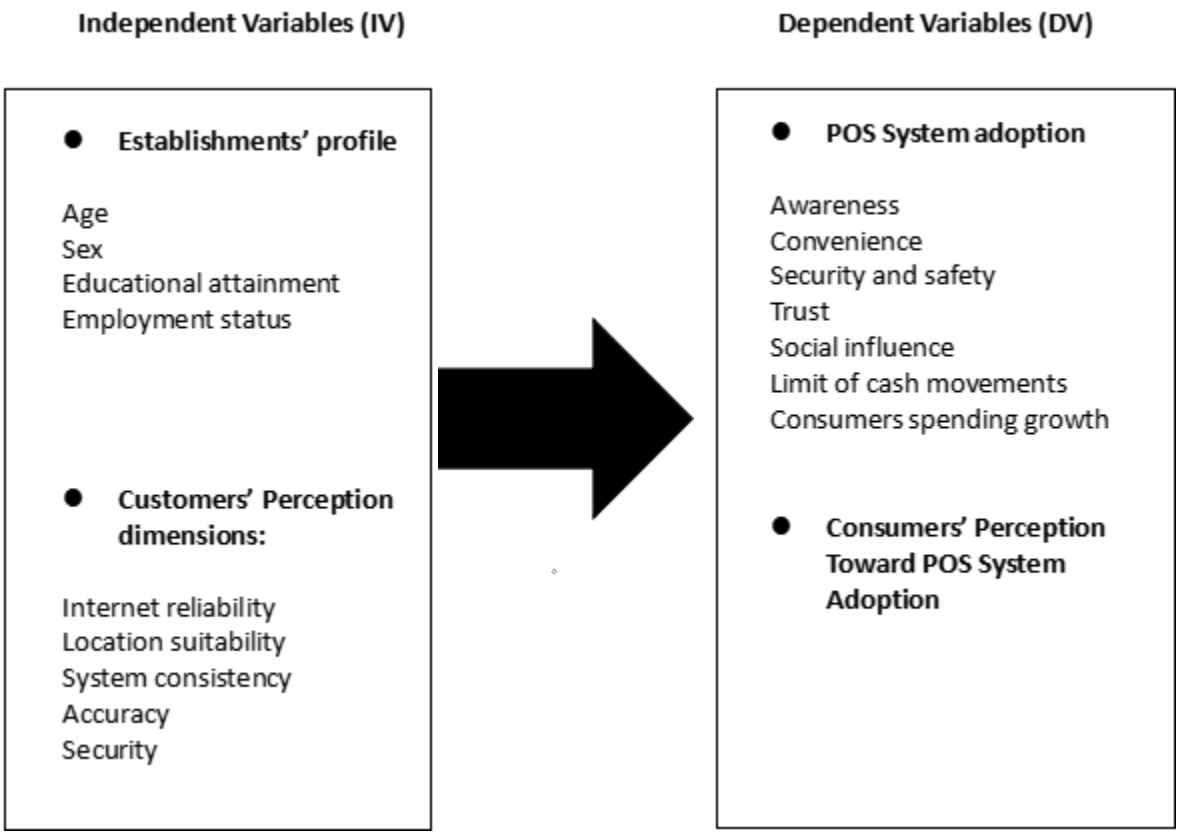
Customers’ perception

Customer perception strongly influences how the adoption of a POS system affects operational effectiveness and efficiency. Röding et al. (2023) classified different types of POS technologies from the customer’s viewpoint, showing that customers’ perceptions of information-oriented POS systems (e.g., how they present product info or transaction details) vary and directly affect their acceptance.

Meanwhile, González-Morera et al. (2023) found that customers visiting Caracol points-of-sale perceive visual merchandising elements positively overall, though aspects such as product distribution, promotions, and ambiance (like music) are seen less favorably, suggesting that POS

implementations reinforcing attractive, well-organized store layouts can boost customer satisfaction.

1.4 Schematic Diagram



1.5 Statement of the Problem/ Objectives of the Study

1. What is the profile of the respondents in terms of:

1.1 Age

1.2 Sex

1.3 Educational Attainment

1.4 Employment status

2. What is the level of impact on POS adoption among establishments in terms of:

2.1 Awareness

2.2 Convenience

2.3 Security and Safety

2.4 Trust

2.5 Social Influence

2.6 Limit of cash movement

2.7 Consumers' spending growth

3. Is there a significant relationship between the profile of the respondents and the level of impact of adopting a POS?

4. How do system-related factors such as internet connectivity, the consistency and placement of the system, transactional accuracy, and data security affect customers' perceptions of POS machine usage within establishments?

1.6 Hypothesis

H_0: There is no significant relationship between the profiles of the respondents and system-related factors and the overall impact of POS system adoption.

H_0 1: There is a significant relationship between the profiles of the respondents and system-related factors and the overall impact of POS system adoption.

1.7 Scope and Limitations

This study will aim to examine the effects of adopting a Point of Sale (POS) system on operational effectiveness and efficiency in the food and beverage (F&B) industry. The research will focus specifically on F&B establishments operating in the municipalities of Buenavista, Nasipit, and Carmen, where the industry has been experiencing notable growth due to rising local economic activity and increasing consumer demand (Beata, et al. 2025). These areas will be selected because they will host a growing number of small- to medium-sized restaurants, cafés, and other F&B businesses, making them suitable sites for studying technology adoption in business operations. Their accessibility and relevance will also ensure the feasibility of accurate data collection.

The study will concentrate on how POS adoption will influence several operational dimensions, including transaction accuracy, service speed, inventory control, cost reduction, and customer satisfaction. Respondents will include employees—cashiers, servers, and managers—who will interact directly with the POS system in their daily tasks, as well as customers, who will not engage with the system directly but will experience its effects through service speed, billing accuracy, and overall service quality. The research will be conducted and analyzed within School Year (SY) 2025–2026.

Despite its defined scope, the study will have several limitations. First, it will cover only a selected sample of F&B outlets within the three municipalities, so findings may not be generalized to other towns, cities, or larger-scale restaurant operations. Second, the respondent groups will be limited to employees and customers, which may result in limited representation from business owners or top management. Third, the study will focus solely on operational outcomes of POS adoption and will not explore broader financial or marketing impacts, such as revenue growth, brand positioning, or long-term strategic benefits. Additionally, customer responses may be constrained by their limited awareness of the POS system's technical functions, as their feedback will primarily reflect service-related experiences. Finally, external factors such as internet connectivity, power supply stability, vendor-related software updates, and regulatory requirements may influence POS performance, but these factors will fall outside the scope of the study.

1.8 Definition of Terms

Accuracy. Refers to the correctness of the POS system in recording, calculating, and processing transactions. It is measured by comparing system records with actual transactions to identify discrepancies in pricing, billing, or inventory.

Age. Refers to the respondent's chronological age (years) as reported in the survey. Age has been considered a demographic variable in this research, affecting merchants' awareness, trust, perceptions of convenience, and adoption of POS systems.

Awareness. Refers to how familiar respondents are with POS systems. It is measured by their self-reported knowledge, recognition of features, and experience using POS transactions.

Consistency. Refers to the ability of the POS system to perform transactions uniformly and predictably without errors. It is measured by tracking transaction success rates, frequency of system errors, or repeated functionality tests over time.

Convenience. Refers to respondents' perceived ease, speed, and convenience of POS transactions in their work or customer experience. Convenience involves quicker checkout, various payment options, and ease of handling transactions.

Customers' Perception Towards POS Machines. Refers to the opinion or attitude of customers regarding the use of POS machines in a business. It is measured through surveys using a Likert scale on statements about convenience, speed, and satisfaction with POS transactions.

Educational Attainment. Refers to the highest level of education a respondent has completed (e.g., high school, college, or postgraduate). It is used in this study as a measure of digital literacy, which can affect how well they adopt and use POS systems.

Employment Status. Refers to the type of work restaurant employees do, such as full-time, part-time, or temporary/contract. It helps assess how their work status affects the POS system's impact on operational efficiency and effectiveness.

Growth in consumer spending. Refers to the perceived or reported growth in customer spending (basket size, purchase frequency, or sales volume) driven by convenience enabled by the POS and lower friction in purchase transactions.

Internet. Refers to the reliability and speed of the internet connection used by the POS system to process transactions. It is measured by the frequency of transaction delays or failures due to internet issues or through ratings from customers and employees on system connectivity.

Limit of Cash Movements. Refers to the level at which the business manages and reduces the number of times it physically handles, stores, or transfers cash in the daily business.

Location. Refers to the physical or virtual accessibility of the POS system within the business environment. It is measured by evaluating the ease of access for customers and employees to POS terminals or by tracking the time taken to complete transactions at different locations.

Operational Effectiveness. Refers to how POS systems help businesses and employees work more accurately and reliably. Examples include improved inventory control, fewer transaction errors, and smarter decision-making.

Security. Refers to the measures implemented to protect the POS system and customer data from unauthorized access, fraud, or breaches. It is measured by the presence of security protocols (e.g., encryption, user authentication), incidence of security breaches, or the perceived safety by customers and staff.

Security and Safety. Refers to how safe respondents feel when using POS systems, especially regarding fraud, mistakes, and data breaches. It is measured by their trust in the system's technical security and their sense of safety while making transactions.

Sex. Refers to the restaurant customer's sex, which is determined by whether the respondent is male or female, as transpired in the demographic profile part of the questionnaire. This variable tests the potential variation in the perceptions of POS awareness, trust, and security.

Social Influence. Refers to the impact of peers, coworkers, managers, and customers on the respondents' use of or dependence on POS systems. It is calculated by the perceived support or normative influence to use and embrace POS technology.

Trust. Refers to how confident restaurant customers feel about using POS systems. It includes trust in the technology, the service providers, and the merchants. It is measured by survey questions about reliability, transparency, and confidence in using POS information.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter will discuss the research design, locale, respondents, and instruments used in the study. It will also explain the ethical standards, data-gathering procedure, and statistical treatments.

2.1 Research Design

The researchers will utilize a quantitative-descriptive correlational research design to find the relationship between Point-of-sale system and Operational Effectiveness and Efficiency. In this design, the study will involve gathering numerical data through survey questionnaires that describe and analyze the perceptions and behaviors of the respondents. The descriptive part will help the researchers in presenting the characteristics of the respondents, while the correlational part will determine if there is a significant relationship between Point-of-sale system and Operational Effectiveness and Efficiency. This design will be appropriate because it will enable the

researcher to collect measurable data and find out how Point-of-sale system may Impact customers' satisfaction in Buenavista, Nasipit, and Carmen.

2.2 Research Materials and Methods

Research Instrument

This study will use two research instruments to gather data from both establishment employees and customers. The first instrument is a modified survey questionnaire adapted from Oyelami et al. (2020), entitled Electronic payment adoption and consumers' spending growth: empirical evidence from Nigeria, designed to assess the effectiveness of POS systems in restaurant operations. The questionnaire is divided into two parts: Part I collects demographic information from establishment employees, including age, sex, educational attainment, and employment status. Part II consists of 27 items measured on a 4-point Likert scale (4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree) and evaluates the impact of POS systems on operational factors such as order accuracy, service speed, and customer satisfaction.

The second instrument is a modified survey questionnaire adapted from Huma et al. (2025), entitled Influence of mobile application service quality and convenience on young customer retention, which focuses on customer perceptions of POS system adoption. Similar to the first instrument, it is divided into two parts. Part I collects demographic information from customers, including age, sex, educational attainment, and employment status. Part II also contains 27 Likert-scale items and assesses customers' experiences and satisfaction with POS systems, including perceived improvements in service efficiency, payment convenience, and overall dining

experience. Together, these two instruments allow the study to capture both the operational perspective of employees and the experiential perspective of customers, providing a comprehensive understanding of the effectiveness and adoption of POS systems in local food and beverage establishments.

2.3 Research Environment, Population, and Sample

The study focuses on the operational impact of Point of Sale (POS) systems within food and beverage (F&B) establishments in the municipalities of Buenavista, Nasipit, and Carmen. These areas represent a growing hub for local commerce, making them ideal environments to study technological adoption in service efficiency.

Recent industry forecasts indicate robust growth in the Philippine food and beverage (F&B) sector, with national food sales projected to increase by approximately 5% and food service sales expected to rise around 10% in 2026. Long-term projections anticipate the foodservice market nearly doubling by 2031, with a compound annual growth rate of about 14%, reflecting strong consumer demand and sector expansion. At the regional level, the economy of Agusan del Norte, which includes the municipalities of Buenavista, Nasipit, and Carmen, grew 6.5% in 2024, with accommodation and food service activities among the fastest-growing segments. This combination of national and local growth highlights the expanding F&B industry in these municipalities, creating a favorable environment for the adoption of technologies such as Point of Sale (POS) systems to enhance operational efficiency and service delivery.

Sample Size Determination

To ensure the study's findings are statistically representative, the researchers utilized Cochran's Formula to determine the sample size for the customer respondents. This formula is particularly advantageous for large populations as it allows for a precise calculation based on desired confidence levels and margins of error.

For the estimated population of 2,140 customers, the initial calculation yielded a sample size of approximately 385. However, because the population is finite, the Cochran's Modified Formula for Finite Populations was applied.

This adjustment resulted in a final required sample of 327 respondents. For the establishment employees, a total enumeration was used due to the small, specific population of 19 direct POS users.

To ensure the relevance and integrity of the data, this study established clear inclusion and exclusion criteria: the included respondents were individuals aged 18 years and above who had prior experience dining in a restaurant that uses a POS system, as well as employees working in establishments that are currently using a POS system and who directly operate or interact with it; excluded from the study were individuals below 18 years of age and employees who do not use or have no experience operating the POS system in their workplace.

Table 1. Establishments' employee

POPULATION				**SAMPLE**					
RESPONDENTS (Employees)	**Male**	**Female**	**Total**	**Male**		**Female**		**Total**	
	F	**F**	**F**	**f**	%	**f**	%	**f**	%
Buenavista									
Roadside		2	2		0.0	2	10.5	2	10.5
Backyard	1		1	1	5.3	0	0.0	1	5.3
Snack break		2	2		0.0	2	10.5	2	10.5
Cafe Aria		1	1		0.0	1	5.3	1	5.3
Aiedee's		1	1		0.0	1	5.3	1	5.3

Nasipit									
Buprs Koi		1	1		0.0	1	5.3	1	5.3
3Js		1	1		0.0	1	5.3	1	5.3
Cafearia		1	1		0.0	1	5.3	1	5.3
Five star	1	1	2	1	5.3	1	5.3	2	10.5
Project Moi		1	1		0.0	1	5.3	1	5.3
Carmen									
Mahigalaon	1	1	2	1	5.3	1	5.3	2	10.5
Painawa		1	1		0.0	1	5.3	1	5.3
Mt. Carmel		1	1		0.0	1	5.3	1	5.3
Zander's food hub		1	1		0.0	1	5.3	1	5.3
Mangroove		1	1		0.0	1	5.3	1	5.3
Total	**3**	**16**	**19**	**3**	**15.9**	**16**	**84.6**	**19**	**100.0**

Table 1 shows the distribution of establishment employees involved in the study, thereby further expanding the number of employees from Buenavista, Nasipit, and Carmen establishments.

Table 2. Customers

POPULATION				SAMPLE					
Restaurant	**Male**	**Female**	**Total**	**Male**		**Female**		**Total**	
	F	**F**	**F**	**f**	%	**f**	%	**f**	%
Buenavista									
Roadside	120	80	200	18	5.5	12	3.7	30	9.2
Backyard	20	30	50	3	0.9	5	1.5	8	2.4
Snack break	250	250	500	38	11.6	38	11.6	76	23.2
Cafe Aria	60	90	150	9	2.8	14	4.3	23	7.0
Aiedee’s	50	70	120	8	2.4	11	3.4	19	5.8

Nasipit									
Buprs Koi	40	60	100	6	1.8	9	2.8	15	4.6
3Js	30	40	70	5	1.5	6	1.8	11	3.4
Cafearia	60	90	150	9	2.8	14	4.3	23	7.0
Five star	100	100	200	15	4.6	15	4.6	30	9.2
Project Moi	91	39	130	14	4.3	6	1.8	20	6.1
Carmen									
Mahigalaon	35	35	70	5	1.5	5	1.5	10	3.1
Painawa	40	60	100	6	1.8	9	2.8	15	4.6
Mt. Carmel	60	40	100	9	2.8	6	1.8	15	4.6
Zander's food hub	50	50	100	8	2.4	8	2.4	16	4.9
Mangroove	50	50	100	8	2.4	8	2.4	16	4.9
Total	**1056**	**1084**	**2140**	**161**	**49.2**	**166**	**50.8**	**327**	**100.0**

Table 2 shows the distribution of restaurant customers involved in the study, thereby expanding the number of employees from the Buenavista, Nasipit, and Carmen restaurants.

2.4 Research Data and Analysis Plan

The researchers will use stratified cluster sampling. The groups (strata) will be based on municipality (Buenavista, Nasipit, Carmen) and type of establishment (fast food, full service, kiosk). Customers and establishments' employee will be randomly chosen from each group, with the number of customers and employee reflecting the size of each type of business.

Statistical Treatment

In presenting, analyzing, and interpreting data, the researcher used the following statistical test, namely.

Frequency count. This tool will be used to answer SOP 1 in determining the profile of the respondents.

Weighted Mean. Will use this tool to answer SOP 2 in determining the level of impact among restaurant customers.

Standard deviation. Will be used to determine the level of agreement or the differences among the restaurants' customer opinions. A low standard deviation means that the responses are nearly centered on the average, thus revealing consensus among the participants.

Correlation analysis. We will apply this to find out the connection between the two variables, mainly the respondents' profiles (age, gender, educational level, and job status) and their opinion on the impact of the POS system on different aspects.

2.5 Research Ethical Standards

A. Protection of Intellectual Property Rights (IPR)

This study will uphold intellectual property rights by ensuring proper acknowledgment of all sources and contributions. All data, literature, and research materials utilized in this study will be accurately cited in accordance with academic standards. Respondents' inputs will remain anonymous, and the researchers' own contributions will be clearly distinguished from those of others, ensuring that the intellectual work of all parties is respected and recognized.

B. Informed Consent

Participants, specifically restaurant customers from Buenavista, Nasipit, and Carmen, will be fully informed about the study before participation. Researchers will conduct a brief orientation session explaining the study's objectives, methodology, and potential benefits and risks. Participants will receive clear instructions on their rights, including the right to decline participation or withdraw at any time without consequence. A written consent form will be provided to ensure participants fully understand their rights and voluntarily agree to take part in the research.

C. Data Privacy and Confidentiality

All personal information, including demographic and identifying data, will be collected strictly with participants' consent and maintained in strict confidentiality. Data will be anonymized to protect participant identities. Only the research team will have access to identification codes, and

all collected information will be securely stored and handled to prevent unauthorized access, disclosure, or misuse.

D. Voluntary Participation and Freedom to Withdraw

Participation in this study is entirely voluntary. Participants will be informed that they may withdraw at any point without facing any negative consequences or penalties. This study involves no animal subjects, ensuring full ethical compliance regarding human-only participation.

E. Minimization of Harm and Risk Management

Researchers will engage with participants respectfully, professionally, and with sensitivity to their well-being. Any potential emotional, psychological, or physical risks will be minimized, and participants will be referred to appropriate support services, such as clinics or the Guidance Office, if necessary. The study will include follow-up measures to confirm that participation did not cause harm or distress.

F. Beneficence and Contribution to Knowledge

This study aims to contribute valuable insights into the impact of the POS system on the restaurant industry. The knowledge gained will benefit both participants and the wider community. Upon request, participants will be provided with a summary of the findings to ensure that the research contributes positively to all stakeholders.

G. Justice and Fair Participant Selection

Participant selection will follow fair and unbiased criteria directly aligned with the study's objectives. Inclusion criteria will consist of individuals who meet the following conditions: (e.g., aged 18–25, that is using the e-payment , and willing to provide informed consent). Exclusion criteria will include individuals who do not meet the study's objectives, such as those outside the target age range, not engaged in social media, or unwilling to participate. Efforts will be made to include participants from diverse backgrounds to ensure equitable representation. No discrimination will be tolerated based on gender, age, ethnicity, socio-economic status, or other personal characteristics.

H. Data Integrity and Accuracy

Data will be collected and analyzed using standardized and scientifically validated methods to ensure reliability and accuracy. The research team will formally request access to respondent lists from the relevant municipalities and will critically review all findings to eliminate bias. Every effort will be made to maintain data integrity throughout the research process.

I. Transparency and Honesty in Reporting

All findings will be reported honestly, accurately, and without any manipulation of data. Academic sources will be properly cited, and AI tools or other technologies will be used responsibly to assist with data analysis and reporting. Any use of such tools will be clearly disclosed to maintain transparency and uphold ethical standards.

J. Use of Patented or Copyrighted Materials

No patented materials will be used in this study. All copyrighted materials will be appropriately cited and utilized in compliance with academic integrity and copyright law. The research will respect the legal and ethical guidelines surrounding intellectual property.

K. Ethical Considerations for Animal and Human Trials

In conducting this study, ethical principles for both human and animal research are carefully observed to ensure respect for participants, minimize harm, and promote scientific integrity. All procedures will adhere to internationally recognized ethical standards, institutional review requirements, and regulatory guidelines.

L. Responsible Use of AI and Other Related Technologies

AI and related technologies will be employed responsibly to enhance data analysis, improve accuracy, and support the research process. Use of such tools will be transparent, ethical, and compliant with academic and legal standards, ensuring that technological assistance does not compromise research integrity.

M. Ethical Clearance and Institutional Approval

This study has received ethical clearance from the Institutional Review Board (IRB) of Saint Michael College of Caraga. This approval confirms that the research adheres to all ethical guidelines, including the protection of respondents, maintenance of data privacy, and respect for intellectual property rights.

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